

Parabolas

Montague Hamilton

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1 Questions

1. **Describe fully and test:** $y = -3(x + 2)^2 + 5$ - **vertex, opening direction, shape and the y-intercept.**

The vertex will occur at $(-2, 5)$, as

$$y = -3((-2) + 2)^2 + 5$$

$$y = -3(0)^2 + 5$$

$$y = 5$$

The opening direction will be downwards, as the y values get inverted by the negative in front of the expression, $-3(x + 2)^2$. The shape will be quite narrow, as for every addition to x, y changes by 3 (shrinks in this case). The y-intercept occurs when $x = 0$, so

$$y = -3((0) + 2)^2 + 5$$

$$y = -3 \cdot 4 + 5$$

$$y = -7$$

The y-intercept occurs at $(0, -7)$